Accounting for the Known Unknowns: A Parametric Framework to Incorporate Systematic Waveform Errors in Gravitational-Wave Parameter Estimation

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The parameter estimation (PE) for gravitational wave (GW) merger events relies on a waveform model calibrated using numerical simulations. Within the Bayesian framework, this waveform model represents the GW signal produced during the merger and is crucial for estimating the likelihood function. However, these waveform models may possess systematic errors that can differ across the parameter space. Addressing these errors in the current data analysis pipeline is an active area of research. We introduce parametrizations for the uncertainties in the amplitude and phase of the reference waveform model. When the error budget in the amplitude and phase of the waveform model, as a function of frequency, is known, it can be used as a prior distribution in the Bayesian framework. We also show that conservative priors can be used to quantify uncertainties in waveform modeling without any knowledge of waveform uncertainty error budgets. Through zero-noise injections and PE recoveries, we demonstrate that even 1%-2% of errors in relative phase to the actual waveform model, for a GW150914-like signal and advanced LIGO detector sensitivity, can introduce biases in the recovered parameters. These biases can be corrected when we account for waveform uncertainties within the PE framework. By analyzing a series of simulated signals from mergers with precessing orbits and recovering them using a non-spinning waveform model, we demonstrate that we can reduce the ratio of systematic errors to statistical errors. This approach allows us to address scenarios where specific physical effects are missing in waveform modeling. The code that implements our parametrization for performing PE is available as a Python package 'pycbc_wferrors_plugin', compatible with the PyCBC open source GW analysis library.

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I. INTRODUCTION

GW signals from compact binary merger events provide us with a unique opportunity to probe these highly energetic events. With the network of advanced GW detectors, including the LIGO detectors in Hanford and Livingston, USA, the Virgo detector in Italy, and the KAGRA detector in Japan, the detection of GW signals has become routine [1–5]. Recent catalogs of GW mergers, compiled from the data analysis of the first three observation runs of the LIGO and Virgo detectors, compile over 90 merger events. These catalogs include the analysis from the LIGO-Virgo-KAGRA (LVK) collaboration [6], as well as analyses conducted by other independent groups [7, 8]. The emphasis of the GW community has now expanded from detection to precision science, such as constraining the theory of general relativity [9], inferring the astrophysical distribution of the population of compact binary mergers [7, 10], inferring Hubble constant [11], constraining the equation of state of a neutron star [12–14], etc.

The detection of GWs and the precision of measurements of source properties relies on the accurate modeling of the signals from the merger of compact binary systems such as binary black holes (BBH), binary neutron stars (BNS), and neutron star-black hole (NSBH). These waveform models have been developed over the years with the help of Post-Newtonian (PN) analytical calculation for the early inspiral part of the signals [15], and they rely on calibration from Numerical relativity (NR) simulations for merger and ringdown regimes [16–18]. The signal's loudness and detector sensitivity determine the width of statistical uncertainties in the measurement, while the inaccuracies in the waveform models could lead to systematic errors. The statistical uncertainties go down as the detectors become more and more sensitive while systematic errors remain the same.

In the coming years, the planned upgrades to existing detectors [3, 19] and the addition of new ones, such as LIGO India [20], are expected to enhance the sensitivity of the detector network. Efforts are currently underway to develop third-generation (3G) detectors, such as the Einstein Telescope (ET) [21, 22] and Cosmic Explorer (CE) [23–25], over the next decade. These 3G detectors are anticipated to be significantly more sensitive—by an order of magnitude—than the current generation of detectors, and they are expected to enhance low-frequency sensitivity. As the GW detectors' sensitivities keep improving, the waveform modeling community tries to keep up with the required accuracy to provide unbiased estimates.

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One important question is how accurate waveform models need to be. One may not want to burden the waveform modeling and NR simulation community with generating waveform models much better than the analysis can resolve [26]. However, studies suggest that we are already reaching the point where we can not ignore the systematic bias in a fraction of loud events. In a recent study, [27] it is shown that the typical biases in the inferred source parameters will be roughly equivalent to the standard deviation for the design sensitivity of current-generation detectors, and these biases could increase significantly for third-generation (3G) detectors. While a small bias in the fraction of events may be irrelevant for some studies, it could be critical for those that depend on accurate estimates of source properties. This is particularly true for inferring population characteristics of BBH mergers [28], estimating the Hubble constant using localization volume, issuing pre-merger localization alerts [29], and conducting tests of general relativity [30, 31], among others.

The impacts of the waveform modeling accuracy on estimating the source parameters have been extensively studied [26–28, 32]. The standard approach to account for waveform systematics is combining the posterior samples from different waveform models by assigning appropriate weights [6, 33, 34] or using hyperparameters to sample over different waveform approximants at the likelihood evaluation level [35–40]. These approaches are expected to marginalize potential differences between different waveform approximants. However, if all the waveform models have similar systematic effects, their difference would not accurately represent the true systematics in each model. Another approach is to marginalize the uncertainties using prior distributions analytically. These priors can be constructed using the Gaussian processes regression technique from a training set of accurate templates [41]. Similar techniques can also be applied to address the waveform systematics arising from calibration with NR simulations [42, 43]. There are also approaches to provide probabilistic models for the waveform, which can be used in sampling with appropriate weights [44]. The authors in [45] marginalize over higher order PN terms to mitigate the systematic errors.

This work presents a general framework that accounts for the uncertainties in waveform modeling through parametric models. The additional parameters in PE analysis relate to the waveform's amplitude and phase errors. When we know the expected amplitude and phase error distribution in the waveform as a function of frequency, we can use them as priors in PE analysis. These errors are expected to be a function of parameter space. In the absence of such knowledge, we can be agnostic and use wider priors to capture any possible deviation. We evaluate this framework by introducing arbitrary deviations into reference waveform models and recovering them through our data analysis pipeline. In addition to correcting for biases, we assess the method's ability to quantify the nature of deviations from the true signal, particularly for the higher signal-to-noise ratio (SNR) "golden" binary systems.

This paper is structured as follows: Section II A reviews current techniques used in waveform model developments and the potential sources of systematic biases they might contain. Section II B visits the state-of-the-art PE techniques and also discusses current methods to deal with waveform systematics. Section III discusses modeling GW strain data from the detector and introduces the parametrizations that we use in this work to account for waveform errors in PE. The following section (section IV) discusses the Fisher matrix approach to account for the systematic bias. We also discuss the limitations where the Fisher matrix approach is no longer valid, and we need to do a full PE analysis. In section V, we present the detailed simulations followed by an injection-recovery campaign to test the validity of this method. We also discuss the scenario where we can use these parametric models to account for the missing physical effects not considered in the waveform model. In the end, we discuss the findings of the work and summarize them in section VII. If the reader is already familiar with the PE techniques and the potential sources of errors, they may skip Section II B. Additionally, if the reader is acquainted with the Fisher matrix approach for correcting systematic biases and understands its limitations in specific scenarios, they can skip Section IV. The reader who is just interested in the methodology and results can read sections III, V, and VII in that order.

II. TECHNICAL BACKGROUND

A. Waveform Modeling and Sources of Uncertainties

The GW signal of compact binary mergers can be predicted by solving Einstein's Equations for the two-body problem in the general theory of relativity. However, due to the complexity of these equations, some form of approximations need to be employed to model binary systems. The most sophisticated models combine several modeling techniques. A review of those techniques and models is beyond the scope of this paper, but we give a brief overview of aspects that are relevant to our discussion of systematic waveform errors. Here we focus on black-hole binaries on quasi-circular orbits, as they are the most frequent source of detected GWs. However, many considerations detailed below apply to more general binary sources as well.

A black hole binary and its emitted GW signal are described by several parameters, such as the BH component masses (m_1, m_2), their spins $(\vec{s}_1, \vec{s}_2)$, luminosity distance (D_L), inclination of binary plane with the line of sight (ι), polarization angle with respect to the detector (ψ), coalescence phase of the binary (ϕ), right ascension (RA) and declination angle (dec) in the sky, and the coalescence time (t_c).

When the binary constituents are widely separated and their velocities are small (compared to the speed of light), the system's evolution can be approximately modeled by a power series expansion. The most commonly used approach is the PN formalism that expands in the system's relative velocity (see [15] for a review). Alternative approaches exist, for example the post-Minkowskian formalism expands in the gravitational constant G [46, 47]. In the highly relativistic regime, where the spacetime curvature is large and velocities are high, these approaches become inaccurate. More concretely, PN expansions are well suited to describe the early inspiral of a black-hole binary, but as the black holes approach each other and their velocities increase, purely PN models become inaccurate and unable to describe the merger and post-merger stages.

While the post-merger ringdown of the remnant black hole can be well described by perturbation theory [48, 49], the merger itself can only be modeled by numerically solving Einstein's Equation. Since its breakthrough [16–18], NR has become an important, well established tool in GW astronomy. However, because of its computational complexity, NR simulation can only cover the late inspiral (typically a few tens of orbits), merger and ringdown of the binary coalescence.

Neither of the above mentioned approaches can model the entire GW signal of interest. PN models become inaccurate towards the merger and do not include any merger or ringdown portion. NR is computationally too expensive and cannot cover hundreds or even thousands of orbits before merger. Therefore, several model approaches have been designed to bridge the gap between numerical and analytical methods [50].

Three model families are regularly employed in the analysis of GW observations [6]. The phenomenological models [51–66] are based on a set of hybrid waveforms that smoothly connect analytically derived inspiral signals with NR data. This data set is then described by PN inspired phenomenological formulae with tune-able coefficients that fit across the binary's parameter space. The resulting model is a set of closed-form expressions that describe the amplitude and phase of the GW signal.

Another approach is based on the effective-one-body (EOB) formalism [67], in which the binary system is mapped to a Hamiltonian formulation of a particle orbiting in an effective potential. By introducing higher-order terms and tune-able coefficients that are fit to waveforms from NR simulations, the resulting EOBNR waveforms can describe the binary coalescence accurately up to the merger. Completed by a suitable ringdown attachment, several models of the EOBNR family have been developed [68–78] and applied to the analysis of GW events.

The third model family is called surrogate models. These models are built by decomposing waveform data from NR simulations into a suitable basis and interpolating between them [79]. NR surrogate models [80–83] are very faithful to the NR simulations they were built from, but they are limited by the length and parameter coverage of NR simulations.

All of the modeling approaches approximate the exact solutions of Einstein's Equation. Therefore, they carry systematic uncertainties that are difficult to quantify in their entirety. Numerical simulations start with approximate initial data and discretize the spacetime. The resulting resolution error is often estimated by providing simulation results of the same system for different resolutions. All modeling approaches described above rely on input from numerical simulations, but they also need to interpolate between different simulations to cover the entire parameter space of interest. This interpolation introduces additional inaccuracies that are often less well quantified. Part of the problem is not only the interpolation (or even extrapolation) of existing NR data. The ansatz used to describe the data carries its own limitations, whether it is data driven (such as Gaussian process regression) or analytically motivated.

In summary, all modeling approaches have to balance representing known NR simulation data as accurately as possible while ensuring a smooth and robust interpolation across the parameter space. The techniques employed to accomplish this have advanced over the years, but they cannot be perfect. Therefore, any model's prediction has to be treated as a "best-guess" waveform for each set of parameters. While the faithfulness to NR data is often very impressive, one must not forget the uncertainties arising from each step of the modeling process.

Here we discuss and test a framework that moves away from the "best-guess" paradigm to accounting for uncertain signal predictions in parameter-estimation analyses.

B. Parameter Estimation

In this section, we review the PE techniques for GW data analysis and potential errors in estimating source properties. Readers who are familiar with this can skip to the next section. Time series strain data $s(t)$ in each detector consists of noise $n(t)$ and may contain a transient signal $h(t)$. It is assumed that when the signal is present, the noise and signal are additive to give the strain data, i.e.

$$s(t) = h(t) + n(t) \tag{1}$$

The noise from the detector is modeled as Gaussian and stationary. In the absence of the signal, the detectors are assumed to contain only Gaussian and stationary noise. The Gaussian and stationary time series noise $n(t)$ can be expressed as [84]

$$\mathbf{n} = \frac{1}{\sqrt{2\pi\Sigma}} \exp\{\frac{1}{2}(n(t_i) - \hat{\mu})^T \Sigma_{ij}^{-1} (n(t_j) - \hat{\mu})\}, \tag{2}$$

where $\mathbf{n}$ is a noise realization which is represented as a vector with discrete time samples $n(t_i)$, Σ is a covariance matrix, and Σ_{ij} represent covariance between i and j time bins, and μ is the expectation value of noise $\mathbf{n}$. The